



VELOCITY AI

STRATEGIC INTELLIGENCE IN THE AI ERA

AN INDUSTRY OUTLOOK

SOFTWOOD SAWMILLING & ENGINEERED WOOD · IRELAND & THE UK

The Next 36 Months

How artificial intelligence resets where value is won in timber processing, and why the mills that move first set a lead that compounds.

NOW

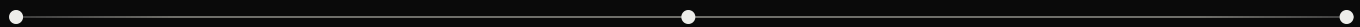
The mill floor

+12 MONTHS

The compliance clock

+24–36 MONTHS

Design & the whole group



The competitive line has moved from owning the best plant to connecting it.

Softwood sawmilling in Ireland and the UK sits at a rare point of alignment. On one side the log resource is tightening, with Irish afforestation slowing and the private sawlog pool thinning, so every mill now competes for a fixed fibre base rather than an expanding one. On the other, demand is rising and becoming regulated, as housing targets and modern methods of construction move timber frame from the edge to the mainstream delivery route.

When fibre is fixed and demand and compliance both rise, two things decide who wins: the value pulled from each log, and the proof carried with each pack. Artificial intelligence is now the mechanism behind both. The plant that leaders have spent decades building, the scanners, graders, kilns and lines, is the foundation. The next increment of advantage sits one layer up, in the data that ties that plant together, buys the log better, proves the carbon, and holds the position as construction moves offsite.

The advantage is no longer only in the plant. It is in the intelligence that runs across it.

6M_m³

Irish roundwood output, still short of what mills and wood energy could take by up to 2.3 million cubic metres.

COFORD · TEAGASC, 2025–26

69%

of scheme homes in the second half of 2025 intended to use modern methods of construction, timber frame in 61%.

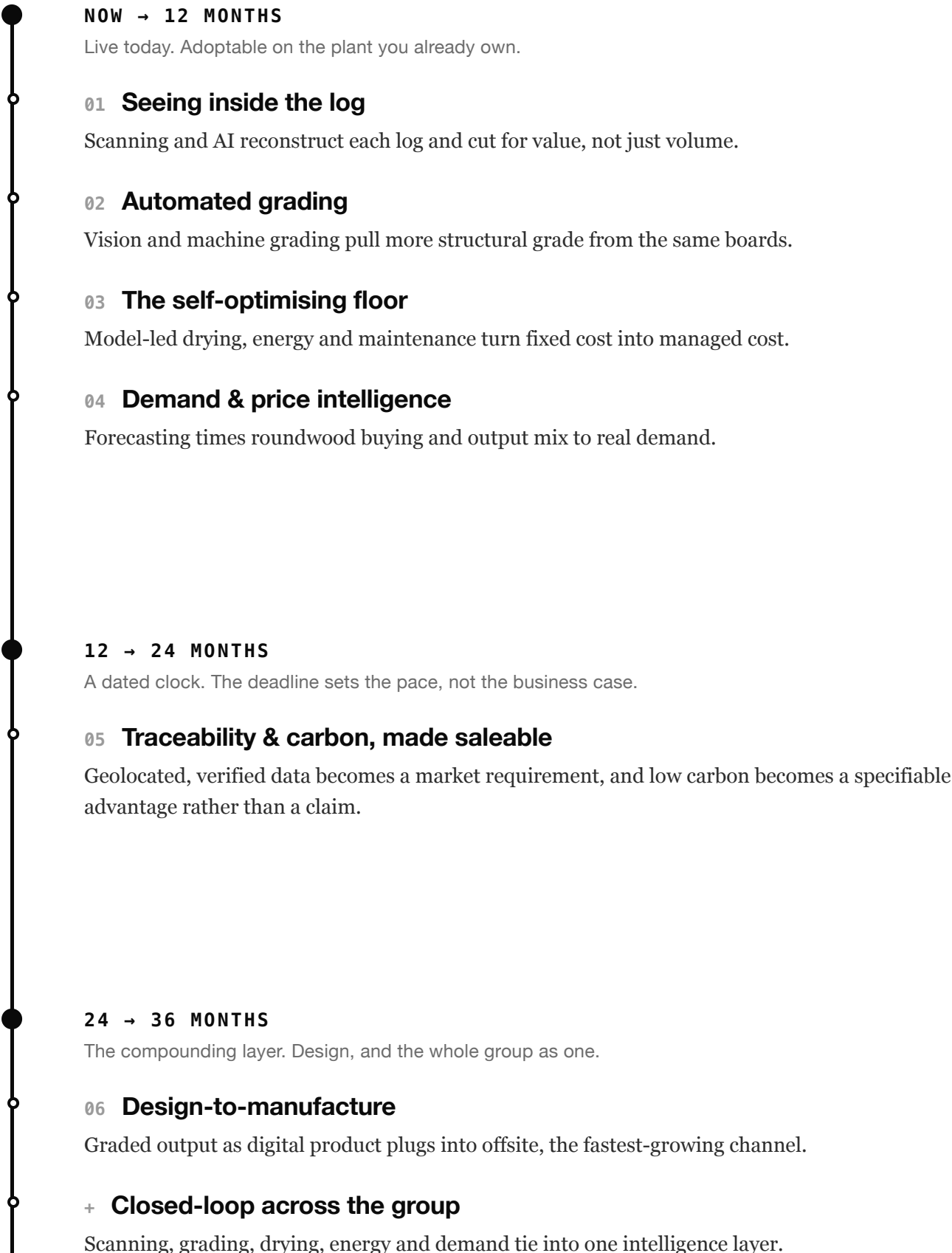
GOV.IE · MMC DATA, 2025

Dec '26

EU deforestation due-diligence applies to large and medium operators from 30 December 2026.

EUROPEAN COMMISSION

Six shifts are already in motion. They do not arrive at once. They sequence, and the sequence is the opportunity: move on the floor now, on compliance next, on the whole group after.



SHIFT 01

Seeing inside the log

The primary breakdown moves from a surface profile to a full internal read of each log, testing many sawing patterns and cutting to the highest value.

The signal. CT and AI reconstruction has shown recovery gains above 11 points in crosscutting and above 4 points overall against centred, horns-down sawing.

SHIFT 02

Automated grading

Camera, laser and X-ray systems read defect, knot and stiffness at line speed, raising average realised grade and giving a defensible record per pack.

The signal. Vision systems are reported reaching up to around 98% defect-detection accuracy, well above manual inspection.

SHIFT 03

The self-optimising floor

Kiln drying, energy and maintenance become model-driven, targeting schedules to predicted moisture and flagging wear before it stops a line.

The signal. Sawmill 4.0 systems pair machine-learning drying with vibration-based predictive maintenance across the process.

SHIFT 04

Demand & price intelligence

Forecasting on order patterns, construction pipelines and price series sets reorder points and times roundwood buying and output mix to demand.

The signal. A softwood market kept volatile by tariff policy makes reading demand and price ahead of the curve a direct margin lever.

SHIFT 05

Traceability & carbon

Geolocation and product-level carbon data move from voluntary to specified, and timber's low-carbon position becomes evidenced and saleable.

The signal. EU deforestation due-diligence applies from 30 December 2026, alongside tightening whole-life carbon rules in public work.

SHIFT 06

Design-to-manufacture

Panel, truss and frame lines take a building model straight to the machine, rewarding suppliers who treat graded output as digital product.

The signal. More than 30% of leading UK housebuilders now run their own panel manufacturing, and offsite is scaling fast.

Every line in this outlook is sourced, and we mark what is firm apart from what is still forming. That discipline is the difference between a view and a verified one, and it is how we work on everything.

THE CLAIM	THE EVIDENCE	CONFIDENCE
Fibre is the binding constraint	Irish roundwood output near 6M m³, short of demand by up to 2.3M m³; first-quarter 2026 afforestation 256 ha against 628 ha a year earlier. COFORD · FARMERS JOURNAL · TEAGASC	● ● ● FIRM
MMC and timber frame are the delivery route	Ireland's MMC Action Plan, 58 actions, June 2025; MMC intended in 69% of scheme homes in H2 2025, timber frame in 61%. GOV.IE · PBCTODAY	● ● ● FIRM
UK housebuilders are internalising panels	More than 30% of leading housebuilders run their own panel manufacturing; Bellway produces all its own wall panels from 2026. Scotland runs near 92% timber frame against about 12% across the rest of the UK. MITEK · STRUCTURAL TIMBER ASSOCIATION	● ● ● FIRM / DEV.
Scanning lifts yield on a fixed log	CT and AI log reconstruction has shown recovery gains above 11 points in crosscutting and above 4 points overall against centred, horns-down sawing. MICROTEC · JOURNAL OF WOOD SCIENCE	● ● ● FIRM
Automated grading raises accuracy	Vision systems are reported reaching up to around 98% defect-detection accuracy, above typical manual inspection. VISION SYSTEMS DESIGN	● ● ● DEVELOPING
Carbon and traceability become regulated	EU deforestation due-diligence from 30 December 2026 for large and medium operators, 30 June 2027 for small; Irish whole-life carbon in larger public projects from September 2025, tightening June 2026. The EU date has moved before, and is currently firm. EUROPEAN COMMISSION · GOV.IE	● ● ● FIRM
The softwood market stays volatile	Cumulative tariff on Canadian softwood moved above 40% after October 2025 measures, keeping global flows and prices in motion, with a gradual recovery expected through the second half of 2026. FASTMARKETS · FARM CREDIT EAST	● ● ● FIRM / DEV.

● ● ● **Firm.** Resolves to a primary or authoritative source, current.

● ● ● **Developing.** Well evidenced in direction, with a figure or date still moving.



Fibre is tightening, demand and regulation are rising, and the mills that move on the floor now, and on compliance and integration next, set a lead that compounds rather than one that erodes.

AI DRIVEN · HUMAN LED

NOW

Move on the mill floor

+12 MONTHS

Meet the compliance clock

+24–36 MONTHS

Connect the whole group

Velocity AI

Operators who build and deploy AI at scale.

Not consultants who leave recommendations behind them.

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